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Deep Learning and Applications

Ching-I Huang

Diffusion Models

$$\mathcal{L} = \mathbb{E}_q \left[\log p_\theta(x_{0:T}) - \log q(x_{1:T}|x_0)\right]$$

$$= \underbrace{\mathbb{E}_q\left[\log p_\theta(x_0 \mid x_1)\right]}_{\text{reconstruction term}} + \mathbb{E}_q\Bigg[\underbrace{\log p(x_T) - \log q(x_T \mid x_0)}_{\text{prior matching term}} + \underbrace{\sum_{t=2}^{T} \log p_\theta(x_{t-1} \mid x_t) - \sum_{t=2}^{T} \log q(x_{t-1} \mid x_t, x_0)}_{\text{Denoising matching terms (for t = 2…T)}}\Bigg]$$

- Reconstruction term
  1. A clean sample $x_0 \sim q(x_0)$ is drawn from the true data distribution.
  2. A slightly noised version $\boxed{x_1 \sim q(x_1 \mid x_0)}$ is generated by applying one step of the forward noising process.
  3. A conditional distribution $p_\theta(x_0 \mid x_1)$,which assigns a probability to every possible clean image x_0 given the noisy input x_1.
  4. The expectation $\mathbb{E}_q[\log p_\theta\,(x_0 \mid x_1)]$ measures how likely the model believes the true clean image $x_0$is, averaged over all noisy samples produced by the forward process q.
  5. This term encourages the model to correctly *explain* clean data even though it only observes the noisy version x_1—that is, to judge how plausible the clean image is given the noisy input.
  6. Because the model is effectively asked to recover the clean data from its noisy counterpart, this component of the loss is referred to as the **reconstruction term**.

Diffusion Models

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- Prior matching term
  1. Forward diffusion:

     As $T \to \infty$, the fully noised variable becomes standard normal: $q(x_T \mid x_0) \approx \mathcal{N}(0, I)$

     $$q(x_t \mid x_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}x_{t-1}, (1-\alpha_t)I).$$

     $$q(x_T \mid x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_T}x_0, (1-\bar{\alpha}_T)I).$$

     $$\bar{\alpha}_T \to 0 \qquad q(x_T \mid x_0) \approx \mathcal{N}(0, I) \quad \text{for all } x_0. \qquad q(x_T) = \int q(x_T \mid x_0)q(x_0)\,dx_0 \approx \mathcal{N}(0, I)$$

  2. The model must adopt the *same* distribution as the starting point of reverse diffusion

     (the reverse process must begin from the same distribution that the forward process ends with.)

     $$p_\theta(x_T) = \mathcal{N}(0, I)$$

  3. Forces the model's prior $p(x_T)$to match the distribution of fully noised data produced by the forward process.

     $$\boxed{p(x_T) \approx q(x_T) \approx \mathcal{N}(0, I)} \qquad \nabla_\theta D_{\text{KL}} = 0 \quad \text{(No trainable parameters)}$$

Diffusion Models

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$$= \mathbb{E}_q\left[\log p_\theta(x_0 \mid x_1)\right] + \mathbb{E}_q\left[\log p(x_T) - \log q(x_T \mid x_0) + \sum_{t=2}^{T} \log p_\theta(x_{t-1} \mid x_t) - \sum_{t=2}^{T} \log q(x_{t-1} \mid x_t, x_0)\right]$$

reconstruction term **prior matching term** **Denoising matching terms (for t = 2…T)**

- Denoising matching term
  1. True Posterior (Forward Diffusion)

     $$\sum_{t=2}^{T} \log q(x_{t-1} \mid x_t, x_0)$$

     If x_0 is the true clean image and the forward process produces x_t, then what should the true distribution of x_{t-1} look like?

  2. Model Posterior (Reverse Diffusion) $x_t \to x_{t-1}$

     $$\sum_{t=2}^{T} \log p_\theta(x_{t-1} \mid x_t)$$

     $$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}(\mu_\theta(x_t, t),\ \Sigma_\theta(t))$$

     The model's predicted Gaussian distribution for denoising one step from the noisy image x_t to obtain x_{t-1}.

  3. The model can transform pure noise into high-quality images step by step.

     Extracting the t-th term

     $$\mathbb{E}_q[\log p_\theta(x_{t-1} \mid x_t) - \log q(x_{t-1} \mid x_t, x_0)] = -D_{\mathrm{KL}}\big(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)\big)$$

     $$\sum_{t=2}^{T} D_{KL}\big(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)\big)$$

     The goal is to make the model's predicted posterior match the true forward posterior as closely as possible—so that at every noise level, the model's denoising distribution aligns perfectly with the true posterior.

Diffusion Models

$$\mathcal{L} = \mathbb{E}_q\left[\log p_\theta(x_{0:T}) - \log q(x_{1:T}|x_0)\right]$$

$$= \mathbb{E}_q\left[\log p_\theta(x_0 \mid x_1)\right] + \mathbb{E}_q\left[\log p(x_T) - \log q(x_T \mid x_0) + \sum_{t=2}^{T} \log p_\theta(x_{t-1} \mid x_t) - \sum_{t=2}^{T} \log q(x_{t-1} \mid x_t, x_0)\right]$$

reconstruction term **prior matching term** **Denoising matching terms (for t = 2…T)**

- Denoising matching term

$$\sum_{t=2}^{T} D_{KL}\left(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)\right)$$

Extracting the t-th Term

$$\begin{cases} D_{KL}\left(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)\right) \\ q(x_{t-1}|x_t, x_0) = \mathcal{N}(\tilde{\mu}_t(x_t, x_0),\ \tilde{\Sigma}_t) \\ p_\theta(x_{t-1}|x_t) = \mathcal{N}(\mu_\theta(x_t, t),\ \Sigma_\theta(t)) \end{cases}$$

$$D_{\mathrm{KL}}\left(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)\right)$$

$$= \frac{1}{2}\left[\log \frac{\det \Sigma_\theta(t)}{\det \Sigma_q(t)} - d + \mathrm{tr}(\Sigma_\theta(t)^{-1}\Sigma_q(t)) + (\mu_\theta(x_t, t) - \mu_q(x_t, x_0))^\top \Sigma_\theta(t)^{-1}(\mu_\theta(x_t, t) - \mu_q(x_t, x_0))\right]$$

The proof is provided on the next page.

Fix the covariance (practical choice) $\Sigma_\theta(t) = \tilde{\Sigma}_t = \sigma_t^2 I$ (known scalar schedule)

$$\begin{cases} \log \frac{\det \Sigma_\theta}{\det \Sigma_q} = 0 \\ \mathrm{tr}(\Sigma_\theta^{-1}\Sigma_q) = d \end{cases} \qquad \log \frac{\det \Sigma_\theta}{\det \Sigma_q} - d + \mathrm{tr}(\Sigma_\theta^{-1}\Sigma_q) = 0.$$

$$D_{\mathrm{KL}}\left(q(x_{t-1} \mid x_t, x_0) \,\|\, p_\theta(x_{t-1} \mid x_t)\right) = \frac{1}{2\sigma_t^2}\left\|\mu_\theta(x_t, t) - \mu_q(x_t, x_0)\right\|_2^2.$$

$$\boxed{\mathcal{L}_{\mathrm{denoise}} = -\sum_{t=2}^{T} \mathbb{E}_{q(x_t|x_0)}\left[\frac{1}{2\sigma_t^2}\left\|\mu_\theta(x_t, t) - \mu_q(x_t, x_0)\right\|_2^2\right]}$$

L2 loss (MSE) An MSE loss between the model mean μ_θ and the true posterior mean μ_q at each timestep.

Diffusion Models

- Denoising matching term

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(\mu_\theta(x_t, t),\ \Sigma_\theta(t))$$

$$p_\theta(x_0|x_1) = \mathcal{N}(x_0; \mu_\theta(x_1), \sigma_1^2 I)$$

$$= \frac{1}{(2\pi\sigma^2)^{D/2}} \exp\left(-\frac{1}{2\sigma^2}\|x-\mu\|^2\right)$$

$$\log p_\theta(x_0|x_1) = \underbrace{\cancel{-\frac{D}{2}\log(2\pi\sigma_1^2)}}_{\text{Constant}} - \frac{1}{2\sigma_1^2}\|x_0 - \mu_\theta(x_1)\|^2$$

Diffusion Models

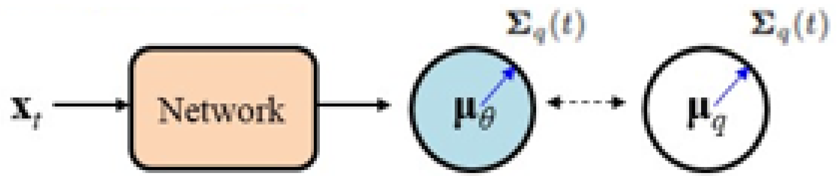
$$\mathcal{L} = \mathbb{E}_q \left[\log p_\theta(x_{0:T}) - \log q(x_{1:T}|x_0)\right]$$

$$= \mathbb{E}_q\left[\log p_\theta(x_0 \mid x_1)\right] + \mathbb{E}_q\left[\log p(x_T) - \log q(x_T \mid x_0) + \sum_{t=2}^{T} \log p_\theta(x_{t-1} \mid x_t) - \sum_{t=2}^{T} \log q(x_{t-1} \mid x_t, x_0)\right]$$

reconstruction term **prior matching term** **Denoising matching terms (for t = 2…T)**

- Denoising matching term

$$\mathcal{L}_{\text{denoise}} = -\sum_{t=2}^{T} \mathbb{E}_{q(x_t|x_0)}\left[\frac{1}{2\sigma_t^2}\left\|\mu_\theta(x_t,t) - \mu_q(x_t,x_0)\right\|_2^2\right]$$



Matching $\mu\theta$ to μ_q is hard to train

1. Train–test mismatch

   The true posterior mean μ_q depends on the clean image x_0 (unavailable during inference)

   $$\mu_q(x_t,x_0) = \tilde{\mu}_t(x_t,x_0) = \frac{\sqrt{\alpha_{t-1}}\beta_t}{1-\bar{\alpha}_t}x_0 + \frac{\sqrt{\alpha_t}(1-\alpha_{t-1})}{1-\bar{\alpha}_t}x_t$$

2. Early timesteps have very small variances (large KL weights)

   $\sigma_t^2 \to 0 \quad \mathcal{L}_{\text{denoise}} \to \infty$

3. Hard regression target

   $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon$

   $t \to \infty \qquad x_t \approx \epsilon$

   $\mu_q(x_t,x_0) \approx A_t x_0 + \text{tiny term}$

   Recovering x_0 from x_t is fundamentally ill-posed for the model.

Diffusion Models

$$\mathcal{L} = \mathbb{E}_q\left[\log p_\theta(x_{0:T}) - \log q(x_{1:T}|x_0)\right]$$

$$= \mathbb{E}_q\Big[\underbrace{\log p_\theta(x_0 \mid x_1)}_{\text{reconstruction term}}\Big] + \mathbb{E}_q\Big[\underbrace{\log p(x_T) - \log q(x_T \mid x_0)}_{\text{prior matching term}} + \underbrace{\sum_{t=2}^{T} \log p_\theta(x_{t-1} \mid x_t) - \sum_{t=2}^{T} \log q(x_{t-1} \mid x_t, x_0)}_{\text{Denoising matching terms (for t = 2…T)}}\Big]$$

- Denoising matching term

By definition,

$$x_t = \sqrt{\alpha_t}\, x_{t-1} + \sqrt{1-\alpha_t}\, \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon$$

the clean image $x_0$with Gaussian noise

ε

$$\epsilon = \frac{x_t - \sqrt{\bar{\alpha}_t}\, x_0}{\sqrt{1-\bar{\alpha}_t}}$$

$$\rightarrow \tilde{\mu}_t(x_t, x_0) = A_t x_t + B_t \epsilon.$$

the true posterior mean depends linearly on the noise ε

Setting

$$\epsilon_\theta(x_t, t) \approx \epsilon$$

Since ε was generated by you, its value is known exactly for that training sample. Thus it can be used as the ground truth label (supervision) in the loss.

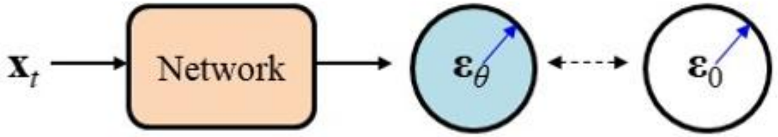
$$\rightarrow \tilde{\mu}_t(x_t, x_0) = \mu_\theta(x_t, t) \iff \epsilon_\theta(x_t, t) = \epsilon$$

$$KL \propto \mathbb{E}_{x_0,t,\epsilon}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\right]$$

A squared-error between the **true noise** and the **predicted noise**

$$\boxed{L_{\text{simple}} = \mathbb{E}_{x_0,\epsilon,t}\|\epsilon - \epsilon_\theta(x_t, t)\|^2}$$

noise prediction objective



- ε is always standard Gaussian, simple and consistent across timesteps.
- $\tilde{\mu}_t$ depends linearly on ε, making reverse diffusion a denoising task.
- Predicting ϵstabilizes training by normalizing the learning signal across timesteps.

Diffusion Models

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- Diffusion models **maximize data likelihood** through the variational ELBO:

$$\log p_\theta(x_0) \geq \mathbb{E}_{q(x_{1:T}|x_0)}\left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T}|x_0)}\right] = \mathbb{E}_q\left[\underbrace{\log p_\theta(x_{0:T})}_{\text{generative model}} - \underbrace{\log q(x_{1:T}|x_0)}_{\text{variational posterior (forward diffusion)}}\right]$$

$$\begin{aligned}\mathcal{L} &= \mathbb{E}_q\left[\log p_\theta(x_{0:T}) - \log q(x_{1:T}|x_0)\right] \\ &= \mathbb{E}_q\left[\cancel{\log p_\theta(x_0 \mid x_1)}\right] + \mathbb{E}_q\left[\cancel{\log p(x_T) - \log q(x_T \mid x_0)} + \sum_{t=2}^{T} \log p_\theta(x_{t-1} \mid x_t) - \sum_{t=2}^{T} \log q(x_{t-1} \mid x_t, x_0)\right]\end{aligned}$$

reconstruction term
(fixed variance $\alpha_1 \approx 1$)

prior matching term
(KL = 0)

Denoising matching terms (for t = 2…T)

KL terms fully determine training quality

- Benefits of ε-Prediction

$$\mathcal{L}_{\text{denoise}} = -\sum_{t=2}^{T} \mathbb{E}_{q(x_t|x_0)}\left[\frac{1}{2\sigma_t^2}\left\|\mu_\theta(x_t, t) - \mu_q(x_t, x_0)\right\|_2^2\right]$$

$$\boxed{L_{\text{simple}} = \mathbb{E}_{x_0,\epsilon,t}\left\|\epsilon - \epsilon_\theta(x_t, t)\right\|^2}$$

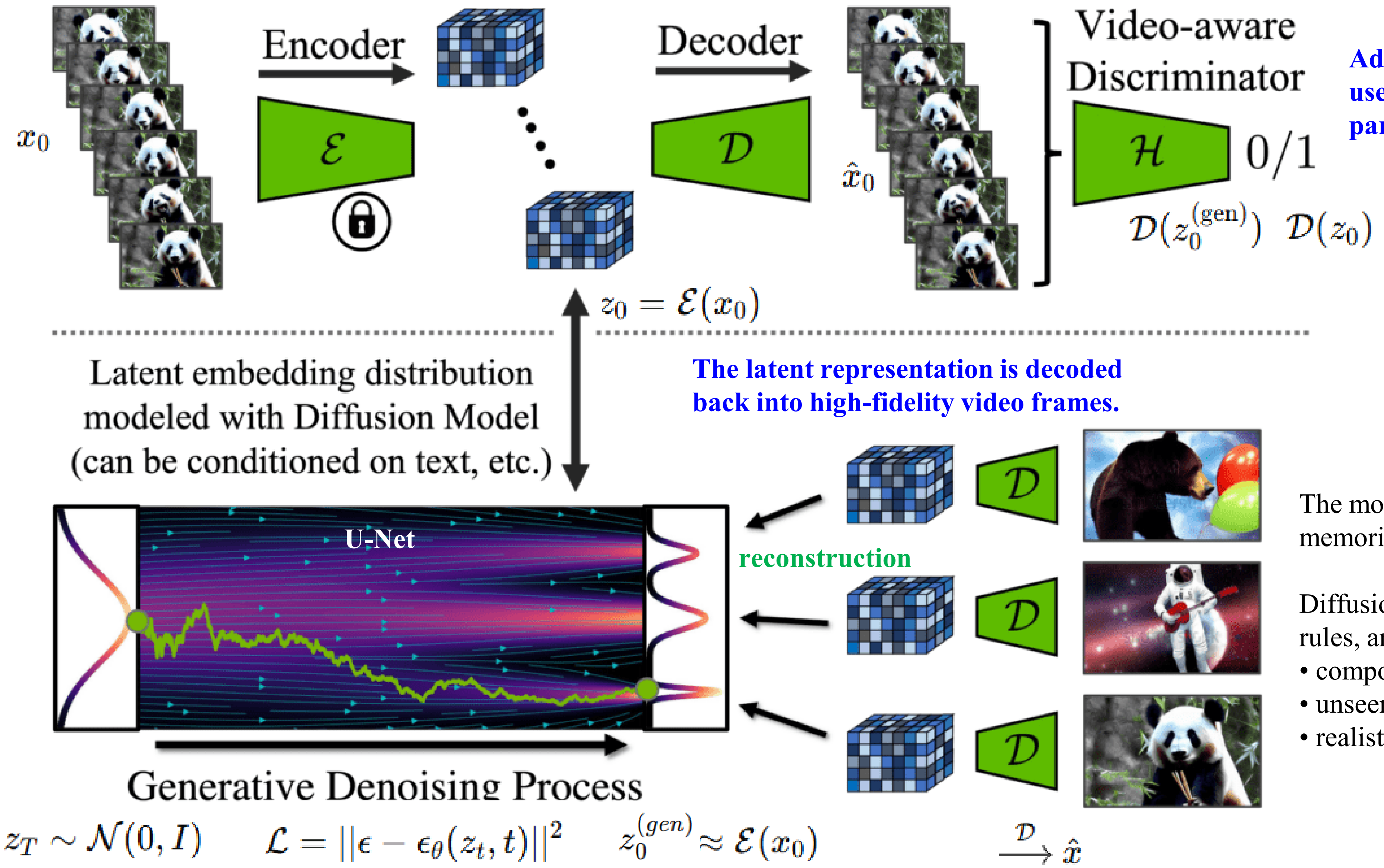
- Natural denoising interpretation
- Learning ε is extremely easy (fixed $\mathcal{N}(0, I)$,identical at all timesteps)
- Full data coverage: The iterative nature of sampling builds in repeated refinement, avoiding collapsed solutions.

Diffusion Models

- Versus GANs
  1. GANs must learn the entire data distribution in one step, resulting in unstable adversarial dynamics.
  2. GANs suffer from mode collapse, missing large portions of the distribution.
  3. Diffusion models provide true likelihood training, stable gradients, and no adversarial game.
  4. The iterative denoising process gives diffusion built-in correction capability, improving robustness and coverage.
- Versus VAEs
  1. VAEs experience **posterior collapse**, where the latent variable becomes unused. The decoder **ignores z entirely** and behaves like a plain unconditional generator
  2. Diffusion models lack an encoder and thus avoid any latent bottleneck or forced compression of information.

Latent Diffusion Models (LDM)

Compresses the input video into latent feature



Adversarial components are optional refinements used only in certain hybrid architectures, not part of the core diffusion training.

The model learns compositional abilities rather than memorizing examples.

Diffusion models learn object structure, composition rules, and visual logic, enabling:

- compositional generalization,
- unseen concept combinations, and
- realistic novel samples on the data manifold.

Generalization in AI

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- **Pre-training**

  Models learn general-purpose representations by training on large-scale datasets.
- **Transfer** (reuse pre-trained weights for downstream tasks)
  1. Fine-tuning: adapt the entire network or selected layers to a new task
  2. Feature extraction: freeze the backbone and train a small classifier on top
- **Generalization** (learn *data distributions*)

  This enables them to perform well not only on the training set,
  but also on new or unseen samples from the same domain.
- **Meta learning**: learn how to learn
- **Few-shot learning**

  The ability to learn new classes or tasks from **very few labeled examples**.
- **Zero-shot learning**

  Perform tasks or recognize new classes **without** seeing any task-specific examples. (semantic or textual relationship.)